FarmRL: Reinforcement Learning for Crop Input Management and Market Timing

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GitHub Repository & Live Dashboard: https://github.com/Harshita772/FarmRL

1. Problem Context

Wheat cultivation across the Indo-Gangetic plains follows a six-month Rabi cycle (November to April). Farmers make recurring fortnightly decisions on nitrogen fertilizer (N), irrigation water (W), and pesticide sprays (P), concluding with a market timing choice at harvest (M): selling immediately or storing grain for 1–2 months. Standard government extension calendars prescribe fixed, calendar-driven inputs regardless of transient weather. In northwest India (Punjab and Haryana), this practice has driven severe groundwater depletion from tubewell over-pumping, topsoil waterlogging during unseasonal winter rains, nitrate leaching into aquifers, and accelerating pesticide resistance. Dynamic, state-contingent input management is therefore essential for sustainable crop production.

2. Constructing the Environment from Empirical Data

To prevent artificial simulation artifacts, we built the **FarmRL** Gymnasium environment by calibrating crop growth, weather sequences, and operating costs directly against published Indian agricultural records.

1. Climate Replay (NASA POWER): The environment replays 39 historical seasons (1985–2023) of daily weather for Ludhiana (Punjab) and Karnal (Haryana), tracking precipitation R_t, min/max temperature, solar radiation, relative humidity, and Penman-Monteith reference evapotranspiration $\mathrm{ET}_{0,t}$. Seasons 1985–2016 (32 seasons) are used for training; 2017–2023 (7 unseen seasons × 30 stochastic replicates) are held out for generalization testing.

2. Yield Response Panel Model (ICRISAT): Harvest yield Y (q/ha) is modeled through a panel regression fitted on 1,719 district-year observations across 13 Indian states from the ICRISAT-TCI database (held-out $R^2 = 0.512$ with state fixed effects). The biophysical response integrates a Mitscherlich nitrogen curve with FAO-33 water deficit response, stage-specific aeration hypoxia $\bar{\sigma}_g^{\mathrm{aer}}$ (FAO-66), pest damage (L_p), and extreme heat-stress days (H_{days}):

$$Y = A_{\mathrm{state}}\, y_{\mathrm{scale}} \left(1 - e^{-\alpha_N (N_{\mathrm{eff}} + N_0)}\right) \left(1 - k_y \left(1 - \frac{\mathrm{ET}_a^{\mathrm{cum}}}{\mathrm{ET}_c^{\mathrm{cum}}}\right)\right) \times \prod_{g=1}^{4} \left(1 - K_{y,g} \bar{\sigma}_g^{\mathrm{aer}}\right) \left(1 - L_p D_{\mathrm{pest}}\right) \left(1 - \alpha_H H_{\mathrm{days}}\right) \tag{1}$$

3. Cost Calibration (CACP): Unit costs are calibrated to official CACP (2024–25) wheat price policy tables: nitrogen fertilizer $c_N = 12.9$ INR/kg, canal and diesel irrigation delivery $c_W = 2.86$ INR/mm, chemical pesticide $c_P = 1{,}183$ INR/dose, and fixed non-input labor and machinery expenses $C_{\mathrm{fixed}} = 29{,}450$ INR/ha. Grain revenue follows the Minimum Support Price (2,425 INR/q) adjusted by Agmarknet market price indices.

4. Four Biophysical Dynamics:

- *Groundwater Drawdown:* Tubewell pumping cost scales directly with water-table lift (25 m base lift in 2024, secular decline of 0.5 m/year plus intra-season drawdown), consuming 0.02725 kWh/mm · ha/m.
- *Waterlogging & Leaching:* Root zone holds 150 mm available water. Moisture exceeding 45 mm above field capacity drains at 20 mm/day, causing root hypoxia (FAO-66) and leaching up to 50% of active soil mineral nitrogen.
- *Pest Resistance:* Pest outbreaks follow humidity shocks; each chemical application degrades subsequent pesticide efficacy by 25%.
- *Storage Market:* Stored grain incurs monthly warehouse fees (10 INR/q/mo), interest (0.58%/mo), and physical grain loss (0.3%/mo) under Ornstein-Uhlenbeck price drift (+1%/mo, 3% volatility).

5. MDP Formalization ($T = 12$ fortnights):

- **State Space ($\mathcal{S}$):** Normalized 13-dimensional continuous vector (extended to 14-D when a next-fortnight rain forecast $\hat{R}_t$ is supplied):

$$s_t = \left[\frac{t}{T},\ m_t,\ \frac{N_t^{\mathrm{soil}}}{100},\ p_t,\ \frac{R_t^{\mathrm{cum}}}{200},\ \frac{N_t^{\mathrm{cum}}}{200},\ \frac{W_t^{\mathrm{cum}}}{500},\ I_s^{\mathrm{price}},\ \frac{R_{t-1}}{50},\ \frac{\mathrm{ET}_{0,t-1}}{100},\ \frac{\mathrm{ET}_{a,t}^{\mathrm{cum}}}{\mathrm{ET}_{c,t}^{\mathrm{cum}}},\ \frac{c_t^{\mathrm{pump}}}{20},\ e_t^{\mathrm{spray}}\right]^{\top} \tag{2}$$

  capturing stage t/T, root moisture m_t, soil nitrogen N_t^{soil}, pest pressure p_t, cumulative inputs, weather lags, water satisfaction, pumping cost, and remaining spray efficacy.
- **Action Space ($\mathcal{A}$):** Factored MultiDiscrete($[3,3,3,3]$) ($|\mathcal{A}| = 81$): nitrogen $N_t \in \{0, 20, 40\}$ kg/ha, irrigation $W_t \in \{0, 37.5, 75\}$ mm, pesticide $P_t \in \{0, 1, 2\}$, and grain storage $M \in \{0, 1, 2\}$ months (masked until harvest step $t = 11$).
- **Reward Function ($\mathcal{R}$):** Farm net profit minus environmental externalities, scaled by $S_R = 10{,}000$ ($r_t = R_t / S_R$):

$$R_t = -\left(c_N N_t + c_W W_t + c_P P_t + C_t^{\mathrm{pump}}(W_t)\right) - \left(\lambda_N [N_{t+1}^{\mathrm{soil}} - 80]_+ + \lambda_P P_t\right) + \mathbb{I}_{[t=11]}\left(\mathrm{Rev}_T - C_{\mathrm{fixed}}\right) \tag{3}$$

  where $\lambda_N = 5$ INR/kg penalizes soil nitrogen leaching above safe levels and $\lambda_P = 100$ INR/dose penalizes excessive chemical usage.

3. Baseline Policies and Fair Benchmarking

To ensure fair evaluation, we benchmark RL against five non-RL policies:

- **No-Input Baseline:** Natural soil fertility without intervention (17.1 q/ha yield, 12,234 INR/ha profit).
- **Random Policy:** Uniform random actions, setting a sanity floor (55,336 INR/ha).
- **Recommended Calendar:** Standard Punjab extension advice (120 kg N across 3 splits, 4 irrigations of 75 mm, 1 spray, harvest sale; 74,649 INR/ha).
- **Untuned Threshold Rule:** Standard textbook moisture and pest thresholds (62,780 INR/ha).
- **Tuned Threshold Rule:** Cutoffs optimized via coordinate grid search exclusively on the 32 training seasons (apply 40 kg N if soil N < 100 kg/ha; irrigate 75 mm if moisture < 10% and 37.5 mm if < 60%; zero spray under calibrated pest loss). Achieves 87,232 **INR/ha** on held-out seasons. A retuned calendar achieves 81,072 INR/ha.
- **Hindsight Foresight Oracle:** Theoretical performance ceiling via cross-entropy plan search endowed with clairvoyance of future daily weather and harvest prices (89,654 **INR/ha**).

4. Experimental Progression, Best Model, and Lessons Learned

Table 1 summarizes held-out benchmarking of all policies. Figures 1–2 compare profit distributions and resource conservation, while Figure 3 and Table 2 detail the 12 systematic ablation studies across 5 seeds:

Table 1: Held-out test performance across 7 unseen Punjab seasons (2017–2023; mean ± std over 5 seeds, 30 replicates/season).

Policy / Configuration	Profit (INR/ha)	% Orac.	Yield	Irrig.	Sprays
Hindsight Oracle (ceiling)	89,654	100.0%	51.2	225	0.00
Tuned Threshold Rule	87,232	97.3%	50.6	241	0.00
PPO (base v4, 5 seeds)	**86,584 ± 153**	**96.6%**	**50.3**	**220**	**0.00**
GRPO (critic-free, $G = 16$)	**86,223 ± 680**	**96.2%**	**50.1**	**232**	**0.01**
CEM (rule parameter search)	84,178 ± 192	93.9%	50.4	238	3.46
Branching Dueling DQN	83,617 ± 1,741	93.3%	49.5	221	0.83
Tuned Calendar Rule	81,072	90.4%	48.4	300	0.00
Recommended Calendar (Govt)	74,649	83.3%	45.8	300	1.00
Untuned Threshold Rule	62,780	70.0%	45.9	236	11.80
Random Policy	55,336	61.7%	44.9	458	11.80
No-Input Baseline	12,234	13.6%	17.1	0	0.00

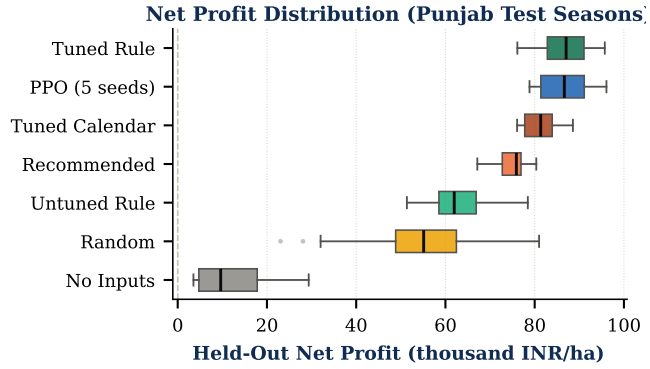


Figure 1: Held-out Punjab net profit distributions (2017–2023): PPO earns +11,935 INR/ha (+16.0%) over the recommended calendar and achieves 96.6% of the hindsight oracle ceiling across 5 random seeds.

1. Baseline Gap and Shapley Split: Base PPO (2×64 MLP, GAE $\lambda = 0.95$, clip $\epsilon = 0.2$) earns 86,584 ± 153 INR/ha, beating recommended extension advice by +11,935 INR/ha (+16.0%) and saving 26.7% irrigation water (220 vs 300 mm). Shapley decomposition across all 24 decision coalitions reveals that 62% of this gain arises from adaptive water timing and 31% from nitrogen splits. PPO trails the tuned rule by −648 INR/ha due to rainfall stochasticity.

2. Optimization Invariance: Sweeping learning rate decay, observation normalization, orthogonal initialization, reward discount $\gamma = 1.0$, and extending training to 1M steps shifted profit by < 0.1%, proving that the gap was informational rather than algorithmic. Branching DQN (83,617) and CEM (84,178) lagged behind.

3. Action Discretization: Introducing a 4-level irrigation grid $\{0, 25, 50, 75\}$ mm paired with a joint $N \times W$ action head lifted PPO to 86,821 ± 315 INR/ha, outperforming its own retuned rule across all seeds.

4. Pest Severity Response: Under severe pest loss ($L_p \geq 0.30$), PPO decisively outperforms the retuned rule across all seeds by +2,601 to +3,827 INR/ha through resistance-aware selective spraying (3.59 vs 6.32 doses at $L_p = 0.35$), preventing chemical resistance buildup.

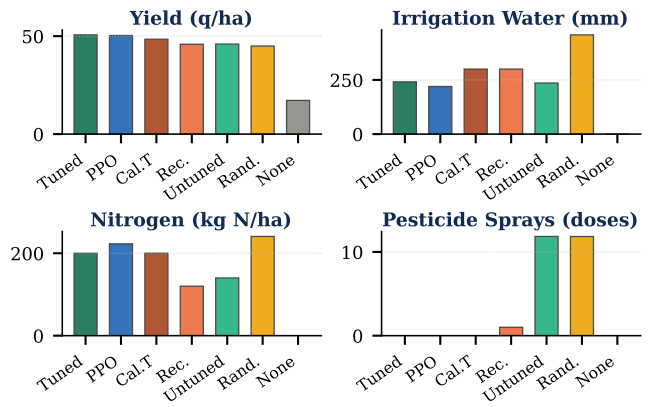


Figure 2: Input resource usage and harvest yield across policies: PPO cuts irrigation by 26.7% (220 mm vs 300 mm) while maintaining 50.3 q/ha yield and eliminating unneeded chemical sprays under calibrated pest pressure.

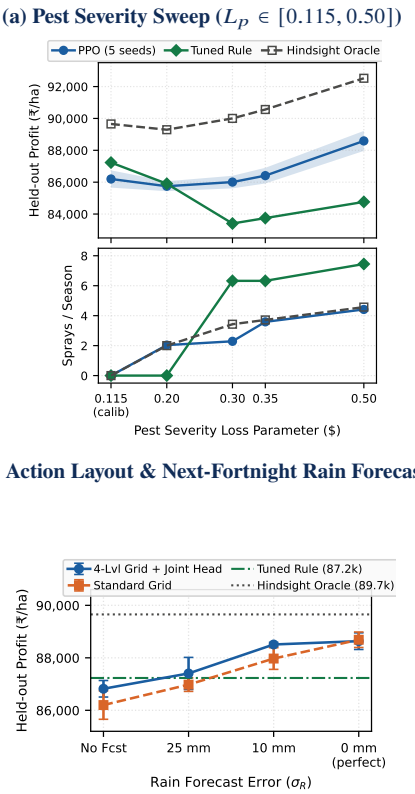


Figure 3: Performance sweeps with zero-overlap legends: (a) For pest loss $L_p \geq 0.30$, PPO beats the retuned rule across all seeds (+2.6k to +3.8k INR/ha) via resistance-aware selective spraying. **(b)** Combining a 4-level water grid, joint head, and a 10 mm rain forecast achieves 88,508 ± 81 INR/ha (98.9% of oracle).

5. Best Model Discovery (Weather Foresight): Oracle decomposition indicates that future weather clairvoyance is worth +5,367 INR/ha vs +188 INR/ha for price knowledge. Supplying a noisy 14-day rainfall forecast with error $\sigma_R = 10$ mm to the 4-level joint agent produced our **best overall model:** 88,508 ± 81 **INR/ha** (**98.9% of oracle ceiling**), beating the tuned rule by +1,963 INR/ha across all 5 seeds.

6. Critic-Free GRPO and Cross-State Transfer: Critic-free GRPO matched PPO (86,223 vs 86,584 INR/ha) with 6% faster runtime. Zero-shot transfer from Punjab to Haryana achieved 74,303± 463 INR/ha (96.2% of Haryana oracle), exceeding Haryana's local tuned rule (73,831 INR/ha) without retraining.

Table 2: Systematic ablation and experiment breakdown across 5 seeds: progression from base PPO to the best foresight model.

Study / Ablation	Intervention / Mechanism	Profit (INR/ha)	Gap to Rule	% Orac.	Win
01: Base PPO	3-lvl grid, separate heads	86,584 ± 153	−648	96.6%	0/5
03: Action Layout	4-lvl water + joint $N \times W$	86,821 ± 315	+621	96.9%	5/5
05: Severe Pest Loss	$L_p = 0.35$ (selective spray)	86,410 ± 492	+2,664	95.4%	5/5
07: Weather Forecast	Standard grid, $\sigma_R = 10$ mm	87,972 ± 412	+586	98.3%	4/5
12: Best Model	**4-lvl grid + $\sigma_R = 10$ mm fcst**	**88,508 ± 81**	**+1,963**	**98.9%**	**5/5**
12: Perfect Weather	4-lvl grid + clairvoyant rain	88,635 ± 317	+1,403	99.1%	5/5
Transfer: Haryana	Zero-shot to Karnal climate	74,303 ± 463	+472	96.2%	5/5

5. Conclusion

FarmRL demonstrates that RL policies discover water-conserving, resistance-mitigating input schedules in complex agro-ecosystems. While tuned static heuristics suffice under predictable conditions, RL policies convincingly outperform them when pest damage is non-trivial and when conditioning on short-term rainfall forecasts. The project includes an interactive 7-page Streamlit decision dashboard.

References

[1] CACP, *Price Policy for Rabi Crops: RMS 2027–28 (Annex 5.6a)*, Govt. of India, 2026. [2] ICRISAT, TCI Cornell, & IFPRI, *District-Level Database for Indian Agriculture (1966–2017)*. [3] J. Doorenbos & A. Kassam, *Yield Response to Water*, FAO Paper 33, 1979; P. Steduto et al., *AquaCrop*, FAO Paper 66, 2012. [4] M. Rodell et al., "Satellite groundwater depletion in India," *Nature*, 460:999–1002, 2009. [5] J. Schulman et al., "PPO Algorithms," *arXiv:1707.06347*, 2017; Z. Shao et al., "DeepSeekMath (GRPO)," *arXiv:2402.03300*, 2024.